

RESEARCH ARTICLE

Minimizing salt pollution in leather manufacturing: A plant-based alternative using *Justicia gendarussa*

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ABSTRACT

Traditional preservation of raw hides and skins typically relies on high concentrations of sodium chloride (NaCl), which contributes to increased salinity and pollution in tannery effluents. This study explores a more sustainable alternative by incorporating *Justicia gendarussa* leaf powder with a significantly reduced amount of salt. Initial trials using varying proportions of the leaf powder (5% to 20%) identified the 15% leaf powder with 10% salt formulation as the most effective based on physical condition and organoleptic evaluation. This optimized combination was applied in a 30-day preservation trial and compared against the standard method using 40% salt. The experimental skins remained stable throughout the preservation period, exhibiting no odor, hair slip, or microbial damage, and retained their softness and flexibility. Analytical results—including moisture content, shrinkage temperature, bacterial count, FTIR spectroscopy, SEM and DSC thermograms—confirmed effective collagen preservation and structural stability. Notably, the soaking liquor from the experimental method showed a substantial reduction in pollution indicators, with TDS, electrical conductivity, and NaCl levels significantly lower than the conventional process. Mechanical testing of the resulting leather showed that the strength and performance characteristics fulfilled or surpassed the standard criteria for shoe upper leather. These findings support the potential of *Justicia gendarussa* as a viable and eco-friendly approach for raw hide preservation, contributing to reduced environmental load and aligning with key sustainable development objectives (SDGs 6 and 12).

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INTRODUCTION

The leather industry stands as one of the oldest and most traditional sectors globally, primarily dependent on the processing of animal hides and skins [1]. This transformation of raw hides/skins into leather involves a sequence of highly resource-intensive operations, relying on raw animal hides, a wide array of chemicals, skilled labor, and complex machinery [2]. The principal raw material, animal hide or skin, is a by-product of the meat processing sector, ensuring a continuous supply stream as long as the meat industry persists [3]. Despite its substantial contribution to national and international markets, the leather industry is recognized for generating significant environmental pollution, especially from centuries-old tanning practices [4]. In response, modern tanneries have emerged with enhanced technological capabilities aimed at better resource utilization and minimizing environmental harm.

As animals are usually slaughtered in abattoirs separate from tanneries, rapid transportation and proper preservation of hides/skins are crucial [2]. The intrinsic nature of rawhide—

comprising approximately 65% moisture, 33% proteins, 0.5% minerals, and 2–6% lipids—renders it highly susceptible to bacterial degradation [3]. Studies have shown that putrefaction can begin within just 5 to 6 hours post-mortem if preventive measures are not applied, and within 8–12 hours, bacteria can penetrate the corium, the functional collagen layer of the skin [5, 6]. Hence, preservation is an essential pre-tanning process that directly influences the final leather quality. Traditionally, preservation is achieved through salting using sodium chloride (40–50% w/w), primarily due to its low cost and ready availability [3]. However, this method poses serious environmental challenges by contributing significantly to the pollution load in tannery effluent, particularly in the form of COD, Total Dissolved Solids (TDS) and chlorides [7]. Excessive salt release into the environment leads to elevated total dissolved solids (TDS) and chloride concentrations in tannery effluents, which, when discharged, degrade surface and groundwater quality. The accumulation of salts in surrounding soils causes soil salinization, which diminishes soil fertility and reduces agricultural productivity in nearby areas. Furthermore, high

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salinity levels inhibit microbial activity in biological treatment systems, thereby reducing the efficiency of effluent treatment plants (ETPs) and increasing the cost and complexity of wastewater management [8, 9]. This practice accounts for more than 50% of COD, 40% of TDS and over 55% of chloride content in composite tannery wastewater [9]. With increasing regulatory scrutiny and rising operational costs of effluent treatment, the leather sector is compelled to explore sustainable alternatives, aligning with SDG 6 (Clean Water and Sanitation) and SDG 12 (Responsible Consumption and Production) [10]. This growing concern has prompted researchers to investigate eco-friendly and reversible preservation methods that minimize salt usage. Another existing approach to minimize salt usage is the direct processing of fresh hides without any curing. While this method completely avoids salt discharge, it poses logistical and microbiological challenges, such as the need for immediate processing and difficulty in long-distance transport, which limit its large-scale applicability [11]. Beside this, alternatives tested include both chemical and physical approaches—such as boric acid, potassium chloride, gamma irradiation, and controlled drying—as well as phyto-based compounds derived from various plants [11-16]. Among these, interest has been steadily increasing in the use of botanical extracts, which are biodegradable, less hazardous, and often exhibit inherent antimicrobial properties suitable for skin preservation [17-23]. Botanical-based approaches, although promising, often show inconsistent preservation efficacy, limited antimicrobial spectra, and poor compatibility with standard leather processing steps. Moreover, there has been minimal exploration of underutilized, regionally abundant plants with rich bioactive profiles, such as *Justicia gendarussa*.

In light of these efforts, the present study explores the use of *Justicia gendarussa* leaves as a plant-based preservation medium in conjunction with a minimal quantity of common salt. *Justicia gendarussa*, locally known as "Jagat Madan" or "Gandoross" in various regions of Bangladesh and South Asia, is a tropical medicinal shrub widely recognized for its therapeutic applications [24-26]. Native to Southeast Asia, *Justicia gendarussa* is rich in several bioactive constituents, including alkaloids, flavonoids, saponins, tannins, and phenolic compounds, each contributing synergistically to its antibacterial efficacy [27-31]. The alkaloids interfere with microbial DNA replication and protein synthesis, leading to inhibition of bacterial growth. Flavonoids and phenolics act as powerful antioxidants and disrupt bacterial cell membranes by forming complexes with extracellular and soluble proteins, ultimately causing cell lysis [32]. Saponins possess surface-active properties that increase membrane permeability, resulting in leakage of intracellular components [33]. Tannins, on the other hand, precipitate microbial proteins and inhibit key enzymes essential for bacterial metabolism [34]. Through these combined mechanisms, *Justicia gendarussa* effectively suppresses bacterial proliferation on hides, thereby preventing putrefaction and reducing the need for excessive salt during preservation. Despite its traditional use in medicine, there has been no reported application of *J. gendarussa* for the preservation of goatskin or rawhide, making this study a novel initiative in this field.

The objective of this work was to examine the effectiveness of *J. gendarussa* in preserving goatskin while reducing the dependency on sodium chloride, thereby lowering the salinity and TDS levels in tannery effluents. The efficacy of preservation was evaluated based on physical and microbiological parameters including moisture retention,

hydrothermal stability, bacterial count, hair slip, odor, surface texture, and any signs of fungal or worm infestation over a 30-day preservation period. A comparative analysis was conducted against control samples preserved using the conventional 40% salt method. Finally, the preserved skins were processed into upper crust leather, and pollution loads during soaking—especially TDS and NaCl content—were assessed to gauge environmental impact. By introducing a *Justicia gendarussa*-based alternative to salt curing, the research contributes to cleaner effluent discharge and sustainable water management (SDG 6). Furthermore, the use of renewable, plant-derived materials promotes eco-friendly and responsible production practices within the leather industry (SDG 12).

MATERIALS AND METHODS

Materials

Industrial-grade sodium chloride (NaCl) and tanning chemicals were procured from suppliers in Savar, Dhaka, Bangladesh. All reagents utilized for analytical procedures were of analytical grade. *Justicia gendarussa* specimens were collected from the Dinajpur, thoroughly washed, sun-dried, pulverized, and subsequently stored under controlled conditions for experimental use. Fresh goatskins were sourced from Kaptan Bazar, Dhaka, and subjected to cleaning and preparation protocols at the Institute of Leather Engineering and Technology for subsequent preservation experiments.

Methods

Collection and preparation of *Justicia Gendarussa* leaf powder

The leaves of *Justicia gendarussa* were collected from Dinajpur, Bangladesh and subsequently air-dried under sunlight for approximately 4 to 5 days. Following complete drying, the leaves were ground using a mechanical grinder to obtain a fine powder (Figure 1). This powdered material was then utilized for the preservation of skins in the present study.



Figure 1. *Justicia Gendarussa* (a) plant leaves and (b) powder

Collection of Goat skin

One piece of freshly flayed goat skin was collected from Kaptan Bazar, Dhaka just immediately after slaughtering to ensure minimal post-mortem changes and immediately subjected to preservation treatment for preliminary analysis

and this preliminary trial was continued for about one week. Another one piece of goat skin was used for this study after the one-week preliminary trial which was collected from the previous mentioned place again following same manner. These pieces were collected with proper care so that no damage was present on the samples. The experiments were conducted under ambient conditions (28–32 °C) without refrigeration to replicate the realistic operating environment of small- and medium-scale tanneries, where cold storage facilities are typically unavailable. The skins were washed with a water flow to remove the dirt and extra bloods for 2-3 times.

Preliminary trial

For the preliminary trial, one piece of freshly flayed skin was collected and cut into four approximate equal parts. The previously prepared leaf powder was used as paste form at different percentages (w/w) based on green weight of goatskin on the flesh part of these parts of skin (Table 1). The paste was prepared by mixing the leaf powder with 10% of NaCl solution in the following four combinations before application (Figure 2).

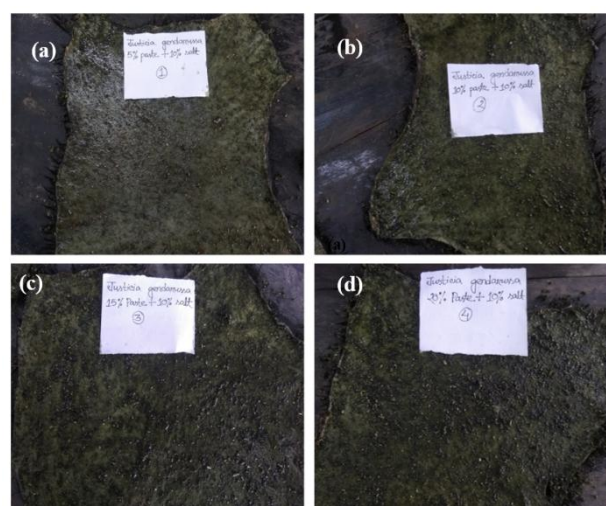


Figure 2. Preliminary trial with (a) S-1 with 5% leaf powder & 10% of NaCl solution, (b) S-2 with 10% leaf powder & 10% of NaCl solution, (c) S-3 with 15% leaf powder & 10% of NaCl solution and (d) S-4 with 20% leaf powder & 10% of NaCl solution

Table 1. Combination of *Justicia gendarussa* leaf powder and 10% of NaCl solution

SL. No	Combination of <i>Justicia gendarussa</i> plant leaf powder and 10% of NaCl solution
S-1	5% of <i>Justicia gendarussa</i> leaf powder + 10% of NaCl solution
S-2	10% of <i>Justicia gendarussa</i> leaf powder + 10% of NaCl solution
S-3	15% of <i>Justicia gendarussa</i> leaf powder + 10% of NaCl solution
S-4	20% of <i>Justicia gendarussa</i> leaf powder + 10% of NaCl solution

Table 2. Organoleptic properties observation during preliminary trial

Combination No	Day	Hair Slip	Odor	Fungal growth	Worm	Physical Feel
S-1	1 st	No	No	No	No	Soft
	3 rd	Slight	No	Slight	No	Medium
	5 th	Medium	Slight	Medium	Slight	Medium
	7 th	Medium to high	Slight to medium	High	Slight	Medium Hard
S-2	1 st	No	No	No	No	Soft
	3 rd	No	No	No	No	Soft
	5 th	Slight	No	No	No	Medium
	7 th	Slight	Slight	Slight	Slight	Medium
S-3	1 st	No	No	No	No	Soft
	3 rd	No	No	No	No	Soft
	5 th	No	No	No	No	Soft
	7 th	No	No	No	No	Soft and flexible
S-4	1 st	No	No	No	No	Soft
	3 rd	No	No	No	No	Soft
	5 th	No	No	No	No	Medium
	7 th	No	No	No	No	Medium Hard

Optimization and selection of the most effective combination

The combinations of *Justicia gendarussa* plant leaf powder and 10% of NaCl solution as curing agents on the skins were assessed periodically for seven days (Table 2). Organoleptic

properties like hair slip, odor, fungal growth, physical feel, etc. were assessed for the enhancement of this study and the best possible combination was chosen for carrying out the next analysis procedures. During these seven days, the 1st, 3rd, 5th and 7th days were to be decided to observe the organoleptic

properties for the quality of the preserved skin. The observation is included in Table 2.

Among the four tested formulations, S-3 demonstrated the most consistent and effective preservation performance, significantly outperforming the other combinations in terms of microbial inhibition, physical stability, and maintenance of hide quality. Throughout the 7-day observation period, S-3 showed no signs of hair slip, odor, fungal growth, or worm infestation, indicating excellent preservation of the skin's structural and biochemical integrity. The absence of hair slip in S-3 suggests that proteolytic bacterial activity was effectively suppressed, preventing the enzymatic degradation of follicular and collagenous proteins[35]. In contrast, S-1 exhibited early signs of degradation, with medium to high levels of hair slip observed by Day 7, along with medium to high fungal growth and slight worm infestation—symptoms indicative of microbial colonization and poor preservation efficacy.

Similarly, S-2 showed moderate performance, delaying degradation until after the 5th day. However, by Day 7, slight signs of hair slip, odor, fungal presence, and worm activity became evident, suggesting a decline in preservation capacity over time. While S-2 initially maintained skin integrity, its limited long-term effectiveness diminishes its suitability for practical application. S-4, on the other hand, maintained good microbial resistance throughout the study, with no visible signs of microbial or parasitic contamination. However, it resulted in a progressive loss of skin softness, culminating in a medium-hard texture by Day 7. This indicates possible moisture loss or early structural stiffening, which could hinder flexibility and workability during downstream leather processing.

In stark contrast, S-3 maintained a consistently soft and flexible texture across all days, indicating superior hydration retention and collagen stabilization[36]. This physical consistency, coupled with the complete absence of microbial deterioration, underscores the formulation's dual functionality: antimicrobial protection and preservation of physical quality. These results suggest that the components of S-3 act synergistically to inhibit microbial growth while preventing dehydration and structural hardening of the hide.

Therefore, in direct comparison, S-3 emerges as the most effective preservation formulation, combining microbial stability with superior physical performance. Its performance far exceeds that of S-1 and S-2, which showed progressive degradation, and even surpasses S-4, which, despite resisting microbial attack, failed to retain the skin's desired physical characteristics. This comprehensive efficacy makes S-3 the optimal choice for short-term goatskin preservation under ambient storage conditions.

Experimental procedure for study

Experimental Procedure was continued with S-3(15% of *Justicia gendarussa* leaf powder + 10% of NaCl solution) as curing agent assessed by the preliminary trial (Figure 3). For this, one piece of freshly flayed goat skin was taken and cut into two equal halves along the backbone. Then another leaf paste was produced with 15% *Justicia gendarussa* leaf powder mixing with 10% salt as it withstands the trial well. This was applied on the flesh side of one half and another half flesh side was applied with 40% NaCl for comparing the result with usual preservation system with our experimental preservation system. This preservation was continued for about 30 Days and whole preservation system was assessed with organoleptic properties, moisture content

determination, total thermal stability determination and total bacteria count.

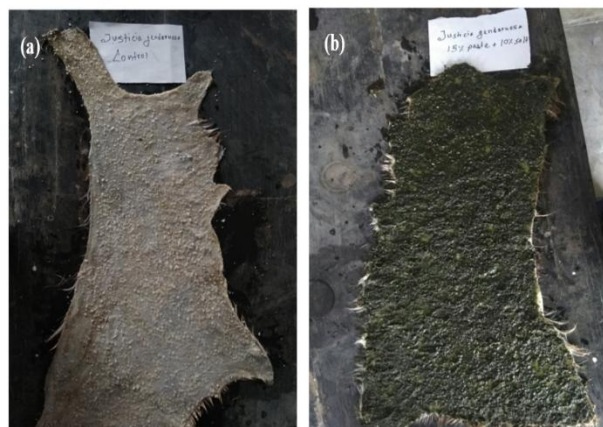


Figure 3. Experimental study with (a) Control sample preserved with 40% salt and,(b) Experimental sample preserved with 15% leaf paste & 10% of NaCl solution

Evaluation of the Preservation Method

Moisture content determination

About 5 g of samples were cut from both the preserved skin samples (Experimental and Control) and then weight was measured. The cut samples taken in a crucible were placed in an oven for about three hours at 105°C. After proper drying, the skins were taken out placed into desiccator for cooling down. Then weights were again measured until a constant weight is achieved. The whole determination procedure and method was conducted according to the given standard [37].

Shrinkage temperature determination

ISO 3380:2015 specifying a method for the determination of the shrinkage temperature of leather up to 100° C was applied for determining shrinkage temperature of the preserved skins [38].

Total bacteria count

The whole method was processed to conduct the bacterial count of the preserved skin samples, about 5 gm of the both skin samples were cut carefully and collected in two separate stomacher bags. These samples were carefully taken to the laminar airflow so that no contamination occurred in the lab during bacterial count test. Then nine times (of the weight of both samples) sterilized water was added to the stomacher bag containing skins. These samples containing bags were shaken at 230 rpm for 30 seconds. After that they were again taken to the laminar airflow. Nine test tubes were taken for each skin samples and filled with 9 ml of water for conducting serial dilution of the skins. Sterilization was performed by taking the test tubes into an autoclave machine and sterilization was done for about 150 minutes. These sterilized test tubes were taken out from autoclave machine and taken to laminar flow again after cooling down them.

1ml of from each stomacher bag was taken with micropipette and poured into both the first test tubes accordingly (expt. and control) containing 9 ml of water in these which resulted in 10^{-2} dilution. These were shaken well after that. Then 1ml from the first test tubes were taken and transferred to the second test tubes respectively forming 10^{-3} dilution. This

process was continued until 10^{-9} dilution is achieved. Petri dishes were already prepared with agar media before which was incubated before and then kept to refrigerator. These were prepared with labelling the sample name (experimental and control) and dilution numbers. The petri dishes were transferred to the laminar flow and the lids were taken out. After that 0.1 ml of the respective solutions were taken out using micropipette from the test tubes and poured them carefully throughout into the petri dishes. Then these petri dishes were gently swirled over by plastic spreader until the solution is properly mixed with the agar without causing any damage to them. Then petri dishes were incubated at 37°C for 48 hours. The next day was reserved for counting the number of colonies formed on the media [39].

Processing of leather

After 30 days of preservation, conventional leather processing method was followed for both the experimental and control goat skins to be manufactured as shoe upper crust leathers. Before processing the leathers, both the skins were indicated with signifying marks so that they could be easily found out after completion of the process. All the pre-tanning, tanning and post-tanning processes were carried out with proper care and awareness following the conventional leather processing techniques [40]. After the conversion of the preserved skins into shoe upper crust leather, the physical properties like tensile strength, grain crack resistance and fiber structure (by scanning electron microscope (SEM)) was assessed appropriately.

Pollution load emission evaluation during leather processing

During leather processing, pollution load was estimated from soaking liquor emerged from the soaking operation from both the experimental and control preserved skin samples. The soaking liquor from both the samples were collected for the determination of Total Dissolved Solid (TDS), Electrical Conductivity (EC) and NaCl content followed by the standard methods [41-44]. TDS was determined gravimetrically, EC was measured using a calibrated conductivity meter, and NaCl concentration was quantified, reported consistently in $\text{g}\cdot\text{L}^{-1}$ (with % w/v in parentheses). Liquor of both experimental and control preserved skin samples were filtered before analysis to remove suspended solids.

Fourier transform infrared (FTIR)

Samples of cured crust leather, approximately 2 mm × 2 mm in size and around 1.0 mm in thickness, were obtained from the standard sampling region of the leather. These samples, representing both experimentally and conventionally preserved hides, were analyzed using a Fourier Transform Infrared (FTIR) spectrometer (Bruker ALPHA II, USA) [45].

Differential scanning calorimetry (DSC)

Thermal stability of the samples was evaluated through Differential Scanning Calorimetry (DSC) using a BK-DSC300L model calorimeter (BIOBASE, China) [45].

Field emission scanning electron microscopy (FESEM)

Leather specimens measuring approximately 2 mm × 2 mm with a thickness near 1.0 mm were obtained from both experimentally treated and conventionally processed crust leathers, following the standardized sampling protocol [30].

For cross-sectional examination, the samples were mounted on aluminum stubs and sputter-coated with a thin platinum layer using an automatic fine coater (JEC 3000FC, JEOL, Japan) to improve electrical conductivity. Imaging was performed with a field emission scanning electron microscope (FESEM) (JSM-7601F, JEOL, Japan) operated at an accelerating voltage of 5 kV and ambient temperature (25 °C), with images captured at various magnifications [45].

Analysis of mechanical attributes of leather

Mechanical properties of the produced crust leathers, i.e, tensile strength, percentage of elongation at break, stitch tear strength, load at grain crack and distension tests were determined by adopting the standard methods IUP-06 norm with Tensile tester (STD 172), DIN 53331 with Tensile tester (STD 172), IUP-09 with Lastometer (STM 104) respectively. Before carrying out the determination, the leather samples were conditioned at $20\pm 2^\circ\text{C}$ temperature and $65\pm 2\%$ relative humidity over a period of 48 hours. All mechanical tests were performed on specimens and cut in both horizontal and vertical directions relative to the backbone to assess the leather properties and each experiment were performed in triplicate. Then samples were taken out from the sampling locations [46].

RESULT AND DISCUSSION

Organoleptic Properties Investigation of the Experimental and Control Skins During Preservation

Raw hides and skins primarily consist of proteins, comprising approximately 30% of their total weight, which makes them particularly susceptible to microbial degradation [47]. The activity of bacteria generates proteolytic enzymes that degrade structural proteins within the skin, leading to the formation of ammonia—a key indicator of the onset of putrefaction [47, 48]. One of the earliest manifestations of this deterioration is hair slip, as bacteria initially target the proteins within hair follicles. Preservation effectiveness is commonly evaluated based on observable parameters such as hair slip, unpleasant odor, the presence of worms, fungal infestation, and changes in the physical texture of the skin [47]. It is quite necessary to observe the organoleptic properties to evaluate the effectiveness of the proposed preservation method comparing with the conventional salt preservation method. Table 3 shows the organoleptic properties of both the experimental and control goat skin samples during the 30 days preservation of the final experiment.

Bacteria cause the production of proteolytic enzymes which cause the putrefaction of skin. These can easily colonize the hair follicles initializing hair slip; the first indication of putrefaction on skin and proteins breakdown to amino acid then to ammonia also implies bad odor or smell [49]. Thus, ultimately causes severe effect on physical feel of skins. That's why, it came out must to assess the efficacy of the proposed preservation method. The comparative investigations of the organoleptic properties during preservation are displayed in Table 3.

From Table 3, it is clearly seen that both the preserved skins; experimental and control didn't face any hair slip or bad odor during 30 days preservation. Both the experimental and control preserved skins exhibit the same degree of physical feel, softness and flexibility without hair slip and putrefaction odor. Both the preserved skins achieved flexibility from the 9th to the 12th day of preservation.

Table 3. Organoleptic properties investigation during 30 days preservation (both expt. and control)

% of Curing agents	Day	Hair slip	Odor	Fungal growth	Worm Development	Physical Feel
Experiment (15% leaf powder+10% salt)	1 st	No	No	No	No	Soft
	3 rd	No	No	No	No	Soft
	6 th	No	No	No	No	Soft
	9 th	No	No	No	No	Soft and Flexible
	12 th	No	No	No	No	Soft and Flexible
	15 th	No	No	No	No	Soft and Flexible
	30 th	No	No	No	No	Soft and Flexible
Control (40% Salt)	1 st	No	No	No	No	Soft
	3 rd	No	No	No	No	Soft
	6 th	No	No	No	No	Soft
	9 th	No	No	No	No	Soft
	12 th	No	No	No	No	Soft and Flexible
	15 th	No	No	No	No	Soft and Flexible
	30 th	No	No	No	No	Soft and Flexible

Effectiveness Assessment and Comparative Evaluation through Different Parameters

Moisture content

Moisture content (%) is an important parameter through which the degree of skin degradation and effectiveness of a preservation method evaluation can be performed. Raw skins containing 60-70% moisture content easily prone to bacterial attack. During the 30 days preservation of the optimized (experimental- using 15% leaf powder+10% salt) and control (40% salt); both the preserved skins were assessed by determining the percentage of moisture content on the 1st, 6th, 9th, 15th and 30th days of the preservation period. The results achieved from determination were displayed in the Figure 4.

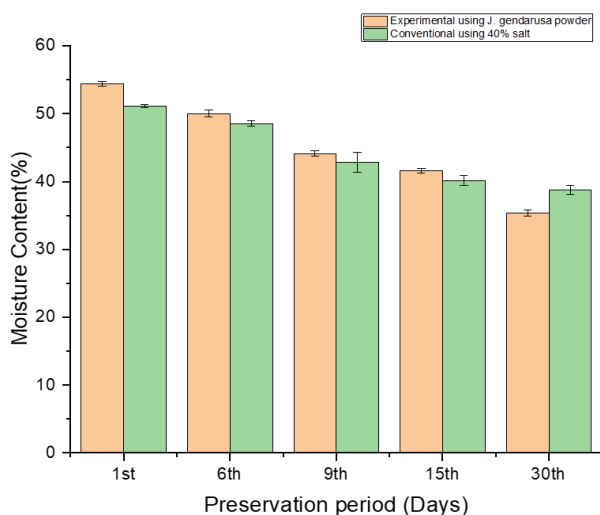
**Figure 4.** Moisture content (%) determination of the preserved experimental and control skins

Figure 4 has been used for making a visual presentation between the moisture content (%) present in the experimental (preserved with 15% leaf powder+10% salt) goat skin with the conventional preservation method during the 30 days preservation period. From Figure 4, it is clearly seen that moisture content (%) of both the samples decreased with the time of 30 days. Though there wasn't any significant variation in moisture content (%), it was slightly higher (3.3%) in experimental on the very first day. The moisture content of experimental decreased to 35.37% on the 30th day which is a little bit (3.38%) lower than controls one. Gradual decrease was observed during 30 days preservation of both sample. Although experimental preservation method showed 6.68% more reduction on moisture content comparing with conventional one, it was higher on the 6th, 9th and 15th day. As there was no sign of putrefaction or skin degradation during preservation, it can be justified as *Justicia gendarussa* leaf showed satisfactory performance on maintaining required moisture of the preserved skins [50].

Shrinkage temperature

Shrinkage Temperature (Hydrothermal Stability) is an important parameter which can investigate the degree of structural stabilization of collagen matrix. For understanding the efficacy of the proposed method, an analysis was made using the collected data presented in the following (Figure 5). The variations on the shrinkage temperature on the experimental and conventional preservation method weren't much noticeable. The shrinkage temperature values for hides treated with *Justicia gendarussa* leaf powder and conventional salt-based preservation showed fluctuations over the preservation period. Notably, the experimental treatment generally exhibited slightly lower values in the early days but converged toward the control values over time. However, it is important to recognize that shrinkage temperature measurements on fresh hides containing hair/wool can be influenced by these natural components and may not accurately reflect structural stability; this test is

conventionally used for evaluating tanned hides. Despite these limitations, the data indicate that *Justicia gendarussa* leaf powder can maintain hydrothermal stability comparable to conventional salt preservation, suggesting its potential to reduce salt usage without adversely affecting the collagen matrix. Further studies employing methods specifically suitable for fresh hides are recommended to validate these observations. So it can be a indication that *Justicia gendarussa* leaf paste can reduce the use of salts in the conventional technique without causing any damage to the structural stability of collagen matrix [51].

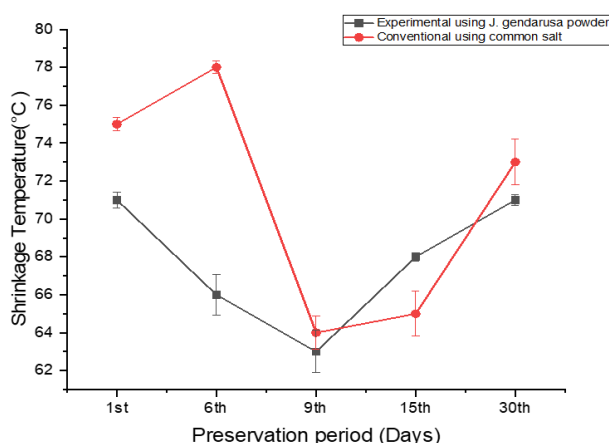


Figure 5. Hydrothermal stability determination of the preserved experimental and control skins

Total bacterial count

Bacterial Count on the preserved goat skins (experimental and control) is one of those important parameters which can depict the performance of the preservation method by investigating whether the method is inhibiting bacterial growth or not. This bacterial count was conducted on both the skins preserved with experimental and control method. The results were shown at the Table 4. It has been visualized that both the experimental and control skins were found to be 3.63×10^9 and 2.37×10^{10} CFU/gm on bacterial count respectively. Bacterial count was decreasing for both the methods through the preservation period was passing out. Specially the decreasing tendency on bacterial count was highly noticed after 15 days on experimental preservation method. The slight increase in total bacterial count observed between days 9 and 15 in both experimental and control skins can be attributed to the natural microbial growth dynamics during the early-mid phase of preservation. Microbial populations can fluctuate due to residual nutrients in the skin matrix, local moisture variations, or adaptation to the preservation environment. Following this transient increase, the bacterial load decreased as the preservative effect of the treatments stabilized over time, limiting further microbial growth. This pattern is consistent with previous reports on microbial dynamics in preserved skins, where temporary fluctuations can occur before a steady decline is achieved. Though both the methods showed efficacy in inhibiting bacterial growth on the 30th day comparing the very first day; experimental preservation method worked well in lowering bacterial count than control method. Therefore, this can be justified as the experimental preservation method (15% leaf powder+10% salt) has antimicrobial activity through which it was successful in inhibiting bacterial population [52, 53].

Table 4. Total bacterial count of the preserved experimental and control skins

Duration	Experiment (CFU /g)	Control (CFU /g)
Day 1	3.63×10^9	2.37×10^{10}
Day 6	3.02×10^9	1.11×10^9
Day 9	1.38×10^9	8.27×10^8
Day 15	3.69×10^8	3.56×10^8
Day 20	2.26×10^8	2.89×10^8
Day 30	1.66×10^7	2.03×10^7

Pollution load generated during soaking operation of (experimental and control) goat skins

The analysis of the soaking liquors from the experimental and control leather preservation methods provides valuable insights into the environmental impact and salt usage efficiency of both systems (shown at Table 5).

The Total Dissolved Solids (TDS) in the experimental soaking liquor was 9836.65 ± 53 mg/L, which is significantly lower than the control value of 11389 ± 324 mg/L. TDS indicates the total concentration of inorganic and organic substances dissolved in the soaking liquor. A lower TDS in the experimental sample suggests reduced salt and chemical leaching, making it a more environmentally friendly alternative. Similarly, Electrical Conductivity (EC), which is directly related to the ionic content of the solution, was also substantially lower in the experimental sample (18.17 ± 0.73 mS) compared to the control (29.25 ± 28 mS). This indicates a decreased ionic load in the wastewater from the experimental treatment, aligning with cleaner production practices. In terms of percentage of sodium chloride (% NaCl), the experimental sample contained 324000 ± 0.86 mg/L, whereas the control sample had a higher concentration of 472000 ± 0.57 mg/L. This shows that the *Justicia gendarussa* powder-based method uses significantly less salt, reducing both chemical input and salinity in the effluent.

The experimental soaking liquor displays markedly lower TDS, EC, and % of NaCl content, suggesting that the use of *Justicia gendarussa* powder is not only effective for leather preservation but also contributes to lower environmental pollution, particularly in reducing saline wastewater discharge. This makes the experimental approach a more sustainable and eco-conscious alternative to conventional salt curing in the leather industry.

Table 5. Pollution load determination of the experimental and control skins during soaking operation

Sample from soaking liquors	Total Dissolved Solid (TDS) (mg/L)	Electrical Conductivity (EC) (mS)	NaCl content (mg/L)
Experiment	9836.65 ± 53	18.17 ± 0.73	324000 ± 0.85
Control	11389 ± 324	29.25 ± 28	472000 ± 0.57

FTIR of leather

The FTIR spectral analysis between conventionally preserved leather using common salt and the experimental sample treated with *Justicia gendarussa* leaf powder highlights key similarities and differences in chemical composition retention (Figure 6). Both samples exhibit broad peaks around 3281.29 cm^{-1} (conventional) and 3303.69 cm^{-1} (leaf powder), indicating the presence of O-H and N-H stretching vibrations, which confirm the preservation of hydrogen-bonded amide and hydroxyl groups in collagen. Similarly, prominent C-H stretching vibrations at 2926.98 cm^{-1} (conventional) and 2924.26 cm^{-1} with an additional peak at 2858.91 cm^{-1} (leaf powder) suggest that aliphatic components remain intact in both treatments, though the sharper appearance in the leaf-treated sample may imply better stabilization. The amide I region, essential for collagen integrity, appears at 1632.25 cm^{-1} in the conventionally preserved sample and at 1635.72 cm^{-1} in the experimental one, confirming effective preservation of the protein backbone in both cases. However, the leaf powder-treated sample shows more defined peaks in the amide II and III regions (1551.09 cm^{-1} , 1447.97 cm^{-1} , and 1236.55 cm^{-1}), compared to the broader and less intense signals at 1543.06 cm^{-1} , 1446 cm^{-1} , and 1234.27 cm^{-1} in the salt-treated leather. Moreover, the fingerprint region exhibits peaks at 1032.49 cm^{-1} (conventional) and 1034.43 cm^{-1} (leaf powder), along with comparable bending vibrations near $664\text{--}658\text{ cm}^{-1}$, indicating the presence of collagen-associated polysaccharide or glycoprotein structures in both samples[45]. Overall, a closer examination reveals that the *Justicia gendarussa*-treated samples consistently show slightly sharper and more defined peaks in regions associated with amide I and II bands, which correspond to collagen backbone vibrations. This suggests a marginally improved preservation of collagen secondary structure compared to the salt-treated samples. Although the differences are not dramatic, they are consistent and reproducible across replicates, supporting the conclusion that *Justicia gendarussa* leaf powder can interact with and stabilize collagen, making it a promising environmentally sustainable alternative.

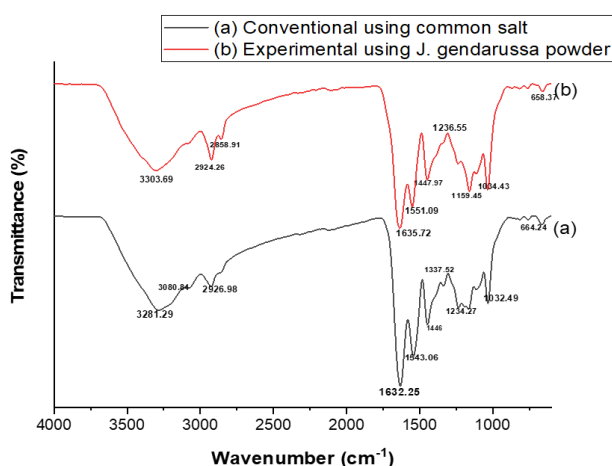


Figure 6. FTIR spectra analysis of experimental and control crust leather

DSC

The DSC thermogram compares the thermal behavior of leather samples processed using a conventional common salt method and an experimental approach employing *Justicia gendarussa* powder. Both samples exhibit a prominent endothermic peak between 90°C and 110°C , corresponding to moisture loss or the evaporation of bound water (Figure 7). This similarity indicates comparable water retention characteristics between the two tanning systems. In the intermediate temperature range (100°C – 250°C), the heat flow remains relatively stable, with minor fluctuations, reflecting polymeric relaxation or molecular reorganization within the collagen matrix. Notably, a significant exothermic transition is observed between 250°C and 310°C in both samples, associated with the thermal degradation or denaturation of collagen fibers[45]. However, the experimental sample shows a broader exothermic response, suggesting a modified decomposition profile due to potential interactions between *J. gendarussa* phytochemicals and the collagen structure. Overall, the comparable thermal transitions indicate that *J. gendarussa*-treated leather demonstrates similar, if not slightly improved, thermal stability relative to the conventionally tanned sample, supporting its viability as a sustainable tanning alternative.

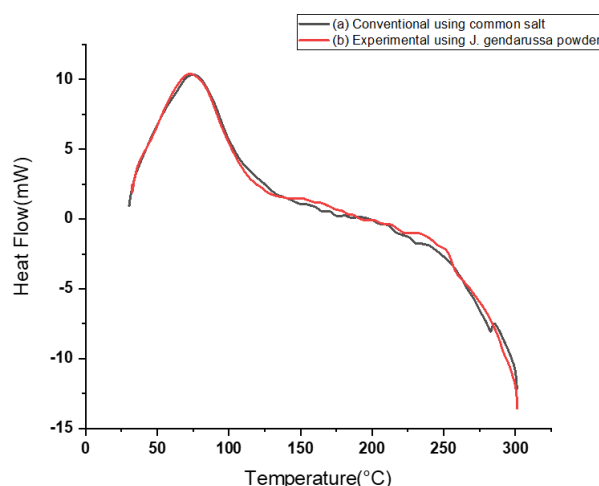


Figure 7. DSC determination of the preserved experimental and control skins

Physical properties evaluation

The experimental leather sample preserved with *Justicia gendarussa* leaf powder demonstrates comparable or superior physical performance relative to the conventionally salted control and clearly meets the standard requirements for shoe upper leather (Table 6). In terms of tensile strength, the experimental sample achieved $245.54 \pm 1.24\text{ kg/cm}^2$, outperforming the control value of $230.20 \pm 1.14\text{ kg/cm}^2$, both of which exceed the minimum required 200 kg/cm^2 , indicating strong fiber cohesion in both treatments. For elongation at break, the control sample recorded 71.37%, slightly exceeding the standard upper limit of 65%, while the experimental sample showed 64.97%, placing it precisely at the permissible upper range, which reflects better dimensional stability in the experimental sample. In stitch tear strength, a key parameter for resistance to mechanical stress at seams, the experimental leather achieved $113.66 \pm 1.09\text{ kg/cm}$, surpassing both the control

(106.81±1.45 kg/cm) and the required minimum of 80–100 kg/cm, suggesting enhanced structural resistance in the leaf-powder-treated leather. Regarding load at grain crack, both experimental (42.21 kg) and control (44.50 kg) samples significantly exceed the minimum 20 kg, confirming strong grain resistance. Lastly, in terms of distension at grain crack, the experimental sample reached 9.67 mm, closely aligned with the control (9.82 mm), and both are well above the 7 mm standard threshold, indicating excellent grain flexibility and durability.

Table 6. Physical properties determination of the experimental and control leather

Physical Properties	Experiment	Control	Requirements for Shoe upper leather
Tensile strength (kg/cm²)	245.54±1.24	230.20±1.14	200
Elongation at break (%)	64.97±0.51	71.37±1.81	40-65
Stitch Tear strength (kg/cm)	113.66±1.09	106.81±1.45	80-100
Load at grain crack (kg)	42.21±0.72	44.50±1.32	20
Distension (mm)	9.67±0.015	9.82±0.005	7

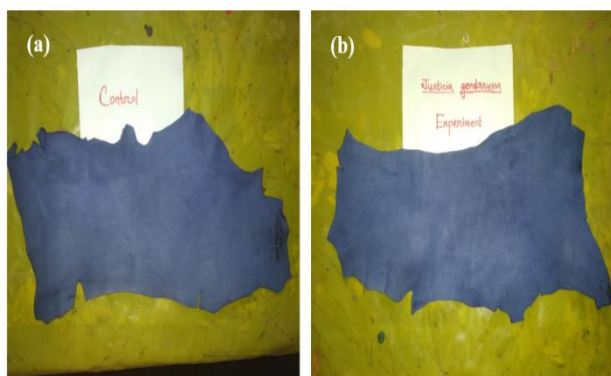


Figure 8. Final look of (a) Control crust leather and (b) Experimental crust leather

The primary objective of the physical properties evaluation was to evaluate whether the experimental preservation method maintained the essential physical properties of leather within the acceptable range for shoe upper requirements, rather than to establish statistical superiority over the conventional method. The results demonstrate that both experimental and control samples meet or exceed the standard specifications, confirming that the alternative preservation method does not compromise leather quality (Figure 8). While the differences between the two sets of data are relatively small, the slightly higher tensile and tear strengths in the experimental sample indicate comparable or marginally improved structural integrity.

FESEM analysis

Field Emission Scanning Electron Microscopy (FESEM) was employed to examine the microstructural characteristics of crust leather samples derived from both the experimental preservation method and the conventional salt-curing process (Figure 9). This analysis aimed to assess the structural integrity of collagen fibers and their resistance to microbial degradation. The FESEM images of the experimental group revealed a compact and coherent fibrous network with well-preserved collagen bundles, indicating effective stabilization of the protein matrix and minimal microbial infiltration. The absence of fiber disintegration or structural collapse suggests that the applied preservation treatment successfully maintained the physical integrity of the leather at the microscopic level [45].

In comparison, the salt-cured control sample also exhibited dense and uniformly arranged fibers without noticeable signs of bacterial attack, thereby reaffirming the established efficacy of salt-curing as a microbial deterrent and structural preservative. The comparative evaluation of both samples indicates that the experimental preservation technique is capable of offering similar, if not enhanced, protection against microbial degradation, while preserving the essential fiber architecture. These findings underscore the potential of the experimental method as a sustainable and effective alternative to traditional salt-curing, particularly in the context of reducing environmental salt load and promoting eco-friendly leather processing practices.

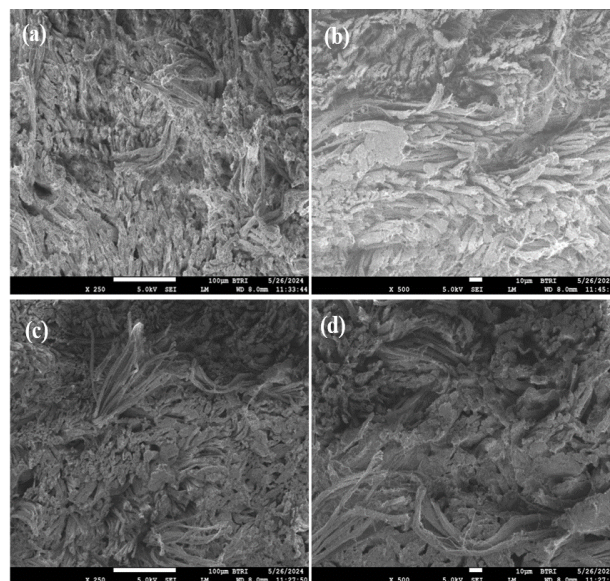


Figure 9. FESEM images of the cross sections from the crust leathers produced from (A)-250X, (B)-500X: Experimentally cured and (C)-250X, (D)-500X conventionally cured leathers

CONCLUSION

The present study establishes that *Justicia gendarussa* leaf powder, when combined with a reduced amount of sodium chloride, can serve as an efficient and environmentally sustainable alternative to the conventional high-salt method for raw skin preservation. Among various formulations tested, the combination of 15% *Justicia gendarussa* powder and 10% NaCl exhibited the most favorable preservation performance over a 30-day storage period. This formulation maintained the natural appearance, softness, and flexibility of

the skin without any indication of microbial growth, hair slip, or putrefaction.

Biophysical and biochemical analyses confirmed the effectiveness of this preservation approach. The experimental skins retained moisture within acceptable limits, and the shrinkage temperature remained within the thermal stability range necessary for further leather processing. Additionally, total bacterial count was significantly lower in the experimental samples, reflecting the antimicrobial efficacy of phytochemicals present in *Justicia gendarussa*. FTIR spectra confirmed the preservation of characteristic collagen peaks, indicating minimal alteration to the molecular structure of skin proteins. DSC analysis revealed consistent thermal degradation patterns between experimental and control samples, further verifying the integrity of the collagen matrix.

Environmental evaluations demonstrated a substantial advantage of the experimental system. The soaking effluent from the *Justicia gendarussa*-salt mixture showed significantly reduced values of total dissolved solids (TDS), electrical conductivity (EC), and NaCl concentration compared to the conventional salt-only method. This suggests that using plant-based preservatives not only lowers salt usage but also contributes to reduced salinity in tannery wastewater—one of the critical challenges facing the leather industry today. Furthermore, the physical testing of finished leathers produced from the preserved skins showed that tensile strength, tear strength, grain crack resistance, and elongation at break met or exceeded standard requirements for shoe upper leather, ensuring the functional viability of the process at an industrial scale.

In conclusion, the use of *Justicia gendarussa* leaf powder as a natural preservative offers multiple advantages: effective microbial inhibition, collagen stabilization, improved environmental sustainability, and satisfactory mechanical performance of the final leather product. While the study primarily focused on preservation efficacy and environmental benefits, the economic feasibility of using *Justicia gendarussa* also appears promising. Being a plant already utilized in traditional medicine, its availability is relatively widespread, and the observations suggest that, when combined with reduced salt, the cost could be competitive with conventional high-salt curing thus can be implemented on detailed cost-benefit analyses for large-scale industrial application. Its required quantity for preservation is relatively low, and the plant is locally available in many tropical regions, making it a feasible option for small- and medium-scale tanneries. This low-salt, plant-based preservation strategy addresses critical environmental concerns associated with salt-intensive curing methods and aligns with the principles of green chemistry and sustainable development, particularly SDG 6 (Clean Water and Sanitation) and SDG 12 (Responsible Consumption and Production). The study recommends further scale-up trials and cost-benefit assessments to evaluate the commercial applicability of this eco-friendly preservation method in the leather industry.

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DATA AVAILABILITY STATEMENT

The authors confirm that the data that supports the findings of this study are available within the article. Raw data that support the finding of this study are available from the corresponding author, upon reasonable request.

CONFLICT OF INTEREST

The authors declare that there is no conflict of interest with any individual, institution, or organization in the preparation, evaluation, or publication of this study.

USE OF AI FOR WRITING ASSISTANCE

Not declared.

ETHICS

There are no ethical issues with the publication of this manuscript.

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